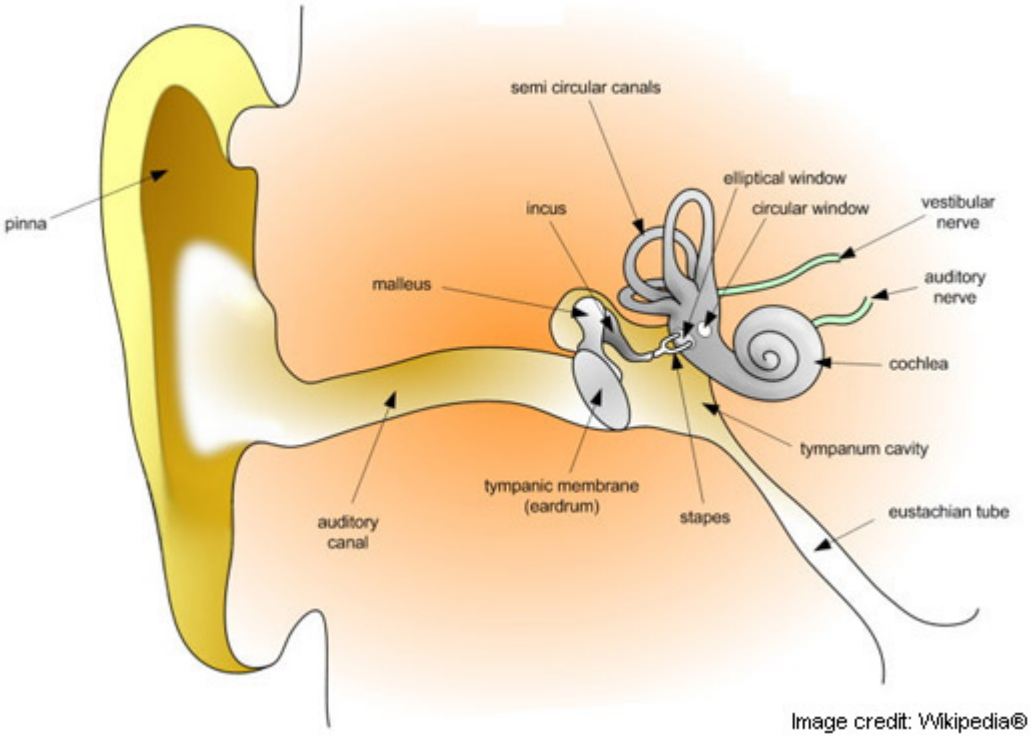
How Hearing Works

Hearing loss is not just an age-related disability; it is affecting people at younger and younger ages. A 2016 study by McMaster University showed that 28% of participants (between 11 – 17 years old) had the early warning signs of hearing loss, in the form of persistent tinnitus (Source (http://dailynews.mcmaster.ca/article/a-disturbing-number-of-teens-show-evidence-of-early-hearing-damage/)).



(http://www.hearingfoundation.ca/wp-content/uploads/2014/04/ear_diagram.jpg)

The Anatomy of the Ear

The outer ear consists of the **pinna** and the **auditory canal**. The auditory canal amplifies frequencies in the range 3 to 12 kHz.

The middle ear includes:

the **eardrum** (tympanic membrane)
ossicles – the three smallest bones in the human body
two **muscle tendons**
two **nerve bundles**: the vestibular (for balance)
the auditory (for hearing)

The inner ear is filled with fluid and includes:

The **cochlea** (for hearing), which each have 15,000 hair cells
The **vestibular apparatus**

The **auditory cortex** is the area of the brain that interprets sounds.

The **eustachian tubes** regulate pressure in the middle ear.

How We Hear

The ear (known as the outer pinna) directs sound waves down the auditory canal to the eardrum (known as the tympanic membrane), causing it to vibrate very slightly. The ossicles include: the hammer (malleus), anvil (incus) and stirrup (stapes). They work together as a lever system to amplify the vibrations, which are then passed along to a smaller vibrating membrane on the surface of the cochlea of the inner ear (called the elliptical window).

The mechanical energy of the sound, now translated to a physical vibration, creates compression waves within the fluid-filled spiral tube of the cochlea, which in turn move the tiny hair cells lining the inside of the cochlea. As the hairs move, nerve cells at their base change this motion into electrical signals that are passed along the auditory nerve to the central auditory processing centres of the brain, where the signals are interpreted as recognizable sound.

To learn more about how hearing works, watch this **video (http://www.rockefeller.edu/labheads/hudspeth/movie06_popup.html)** produced by Rockefeller University.

Decibels – How We Measure Sound

dB (SPL)	Source (with distance)
250	Inside of tornado; conventional or nuclear bomb explosion at 5 m (meters)
180	Rocket engine at 30 m; Krakatoa explosion at 160 km in air
150	Jet engine at 30 m
140	Rifle being fired at 1 m
130	Threshold of pain; train horn at 10 m
120	Rock concert; jet aircraft taking off at 100 m
110	Accelerating motorcycle at 5 m; chainsaw at 1 m
100	Jackhammer at 2 m
90	Loud factory; heavy truck at 1 m
80	Vacuum cleaner at 1m; curbside of busy street
70	Busy traffic at 5 m
60	Office or restaurant inside
50	Quiet restaurant inside
40	Residential area at night
30	Theatre, no talking
10	Human breathing at 3 m
0	Threshold of human hearing (with healthy ears)

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